



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Third Semester – Fall 2022

Paper: Mathematics A-III

Course Code: MATH-201

Roll No. 056519

Time: 3 Hrs.

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following:

(6x5=30)

1. All vector spaces must be abelian group under multiplication. Justify your answer with example.
2. Show that any linearly independent set of vectors of any vector space can be extended to a basis for V .
3. Let V be a vector space over any field and $W \subseteq V$. Then W is subspace of V or not? Justify your answer with example.
4. Define Orthogonal matrices and similar matrices.
5. Define Linear independence and linear dependence of vectors in vector space.
6. Define determinant, cofactors and minors of square matrix.

Q.2. Solve the following:

(5x6=30)

- i) Let A and B be two matrices of order 6 such that $\det(AB^2) = 72$ and $\det(A^2B^2) = 144$.
Find $\det(A)$, $\det(2A)$ and $\det(A^6B^6)$
- ii) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined:
 $T(x_1, x_2, x_3) = (x_1 - x_2, x_3)$. Is T linear? Moreover, find $N(T)$ and $R(T)$
- iii) Find solution of the given system by Gauss-Jordan Elimination method.
 $2x_1 - x_2 - x_3 = 4$; $3x_1 - 2x_3 + 4x_2 = 11$; $3x_1 - 2x_2 + 4x_3 = 11$
- iv) Find Eigen values and corresponding Eigen vector of the matrix $A = \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix}$.
- v) What condition must a, b, c and d satisfy so that the following matrices are linearly independent in M_{22} $\begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$, $\begin{bmatrix} 2 & 3 \\ -2 & 1 \end{bmatrix}$ and $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

PUACP

Orthogonal matrix

8 The square matrix A is said to be an orthogonal matrix

if $A^{-1} = A^t$

i.e., $AA^t = I = A^tA$

Date: 30 Mar, 2024.

Algebra

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Imp topic

Similar Matrices:-

If A and B are two $n \times n$ matrices over $\mathbb{R}$ then A is said to be similar to B if there exists a non-singular matrix P such that

$$B = P^{-1}AP$$

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↳ Linearly Independent

such that

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$$

$$0 = 0$$

$$0v = 0$$

implies that

$$a_1 = 0, a_2 = 0, \dots, a_n = 0$$

Linearly dependent

The vectors $v_1, v_2, \dots, v_n$ are said

to be linearly dependent if there

exists scalars $a_1, a_2, \dots, a_n$

$$x_1 = 2x_2$$

$$1 \cdot x_1 - 2x_2 = 0$$

$$1 \neq 0, -2 \neq 0$$

such that

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$$

implies that $a_i \neq 0, a_2 \neq 0, \dots, a_n \neq 0$

(any one) for some $i = 1, 2, 3, \dots, n$.

Definitions

(i) Minor of an element

Let $A = [a_{ij}]$ be a square matrix,

The minor of an element a_{ij} , denoted by M_{ij} , is equal to the determinant of matrix obtained after deleting the i th row and j th column of matrix A

e.g. if $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 5 & 6 \\ 0 & 1 & 2 \end{bmatrix}$

then $M_3 = \begin{vmatrix} 4 & 5 \\ 0 & 1 \end{vmatrix}$

$$= 4 - 0$$

$$M_3 = 4$$

NOTE

$i = \text{row}$

$j = \text{column}$

(ii) Co-factor of an element

Let $A = [a_{ij}]$ be a square matrix,

then co-factor of an element a_{ij} , denoted by A_{ij} , is defined by

$$A_{ij} = (-1)^{i+j} M_{ij}$$

e.g if

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 4 & 5 & 6 \\ 0 & 1 & 2 \end{pmatrix}$$

then

$$A_{13} = (-1)^{1+3} M_{13}$$

$$= (-1)^4 \begin{vmatrix} 4 & 5 \\ 0 & 1 \end{vmatrix}$$

$$= (1)(4-0)$$

$$A_{13} = 4$$

ii) Determinant

Let $A = [a_{ij}]$ be a square matrix, then determinant of matrix A , denoted by $|A|$ or $\det(A)$, is equal to the sum of the product of elements of a row (or column) and their co-factors

e.g

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Expanding by R_1

$$|A| = a_{11} A_{11} + a_{12} A_{12} + a_{13} A_{13}$$

$$= a_{11} (-1)^{1+1} M_{11} + a_{12} (-1)^{1+2} M_{12} + a_{13} (-1)^{1+3} M_{13}$$

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$$= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$